

VEHICLE SALES

Data Analysis Report

Prepared by

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April 2026

Dataset: 558,837 Records

Tool: Python · Plotly · Seaborn

558,837

Total Records

17

Feature Columns

~98%

Data Quality

01 Data Overview & Cleaning

Dataset at a Glance

558,837

Total Records
Raw dataset rows

548,400

After Cleaning
~98% retained

1982–2015

Year Range
After outlier capping

3.43 / 5

Avg Condition
Normalized score

68,320 mi

Avg Odometer
Before capping

17

Columns
Feature variables

Dataset Description

This dataset is a comprehensive record of car auctions and sales. Each row captures a single vehicle transaction — including the car's specifications, physical condition, pricing benchmark, actual sale price, and the date and location of sale. The breadth of the data makes it well-suited for market analysis, pricing strategy, and demand forecasting.

Column Reference

Column	Description
Id	Unique identifier for each transaction record.
Year	The vehicle's model year — a key driver of perceived value and demand.
Make	Manufacturer brand (e.g., Ford, Chevrolet, Nissan, BMW).
Model	Specific model name within the brand (e.g., F-150, Silverado, Altima).
Trim	Feature and amenity tier — determines equipment level within a model line.
Body	Exterior configuration: Sedan, SUV, Pickup Truck, Coupe, etc.
Transmission	Drivetrain type — automatic or manual gearbox.
VIN	Vehicle Identification Number — globally unique per vehicle.
State	U.S. state in which the transaction took place.
ConditionValue	Condition score on a 1–5 scale; reflects physical and mechanical state.
Odometer	Total miles driven — one of the primary depreciation indicators.
Color / Interior	Exterior paint color and cabin interior color.

Column	Description
Seller	The entity listing the vehicle — dealer, fleet company, or private party.
MMR	Manheim Market Report value — the industry benchmark for fair market price.
SellingPrice	The final price the buyer actually paid to acquire the vehicle.
SaleDate	Date the transaction was completed.

Data Cleaning Approach

The raw data contained missing values, structural inconsistencies, and physically impossible entries. A rigorous cleaning pipeline was applied — handling nulls through statistical imputation, capping outliers at defensible thresholds, and standardizing categorical fields. Every correction was designed to preserve the integrity of real market behavior, ensuring the final analysis reflects actual conditions rather than artifacts of data quality issues.

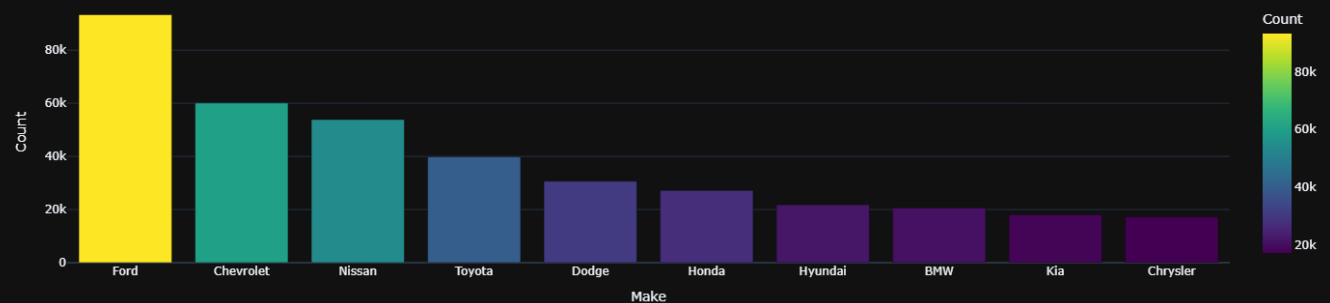
02 Key Insights & Findings

Top 10 Years Models



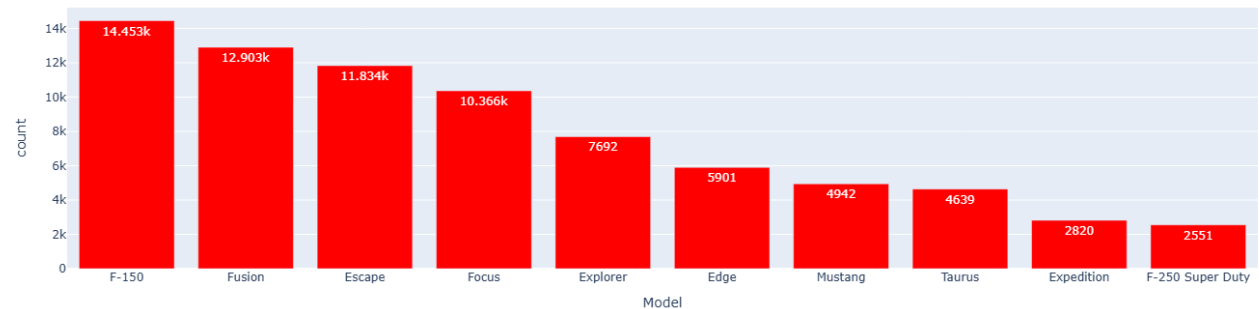
Insight: The market heavily favors newer vehicles — models from 2013 to 2015 dominate sales, confirming that buyers consistently prioritize modern cars over older inventory.

Top 10 Makes



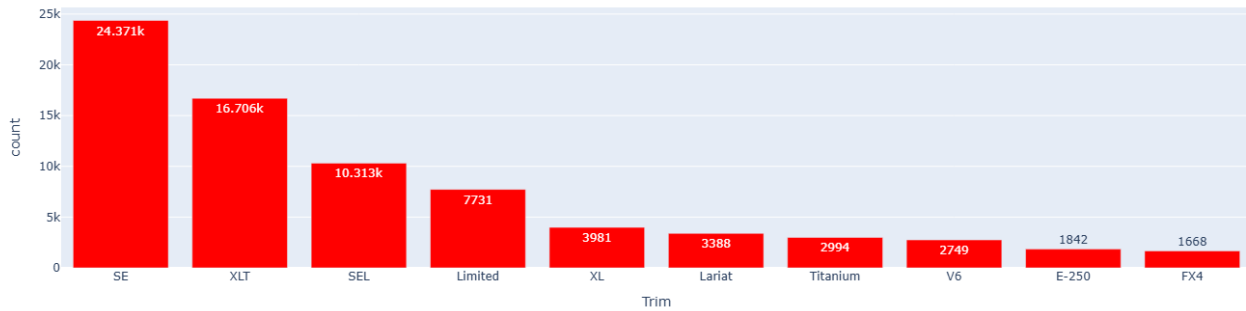
Insight: Ford, Chevrolet, and Nissan lead all brands by volume. Any sales strategy centered on these three manufacturers benefits from the largest and most active buyer base.

Ford Models



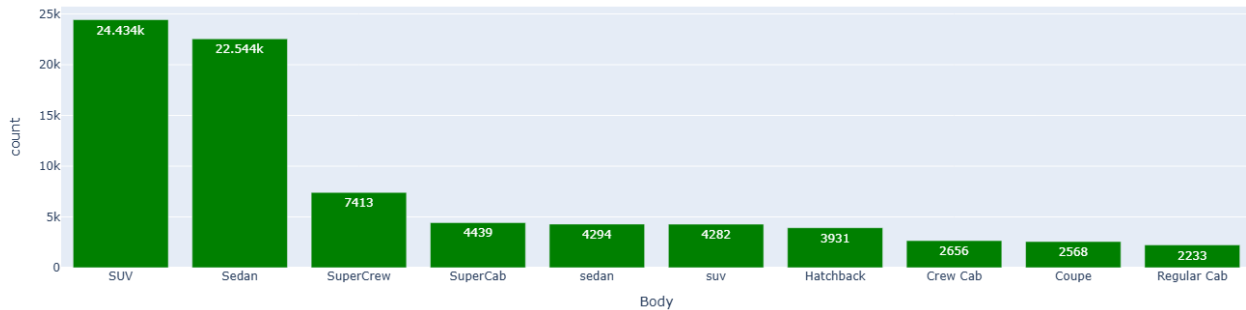
Insight: The F-150 stands alone as Ford's bestseller by a wide margin. For resale, it represents the lowest risk and fastest turnaround among all Ford inventory.

Ford Trim



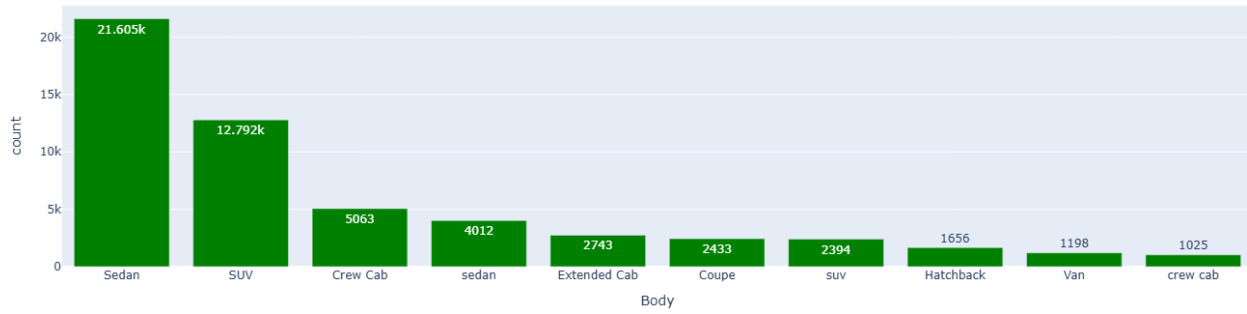
Insight: Mid-tier trims such as XLT dominate demand. Premium trims attract fewer buyers and sit on the market longer — mid-range is where the volume lives.

Ford Body

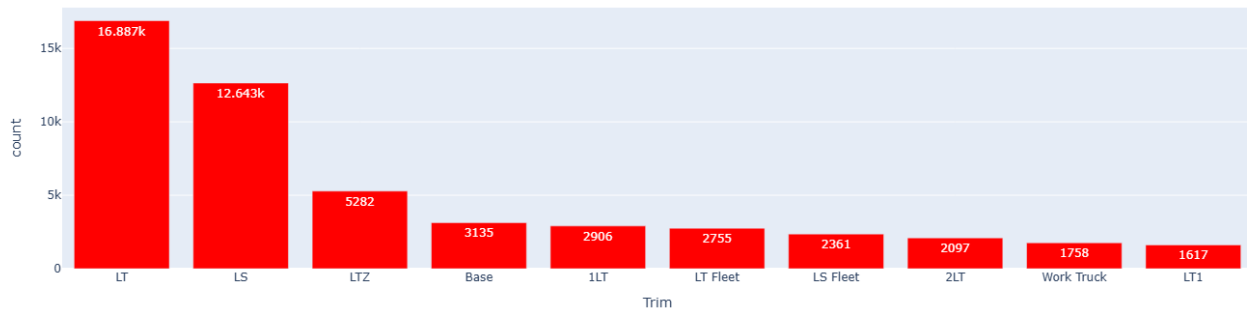


Insight: Pickup Trucks are Ford's defining strength in this dataset. SUVs follow closely. Sedans and smaller body types represent only a minor share of Ford sales.

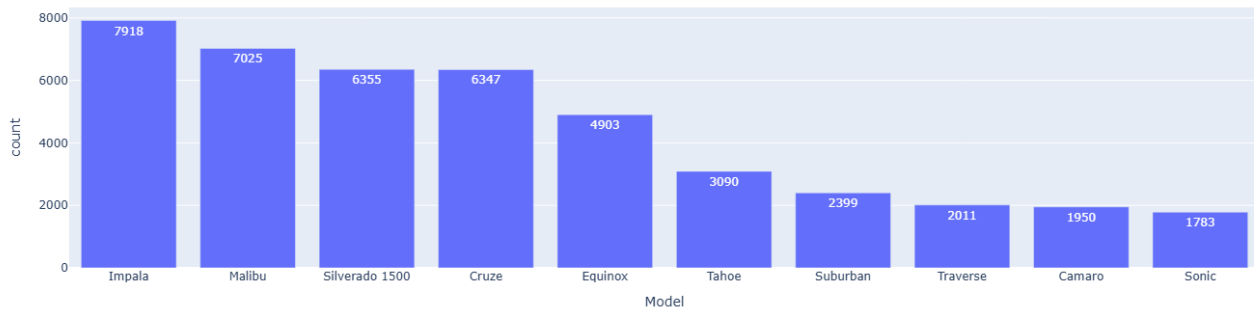
Chevrolet Body



Chevrolet Trim

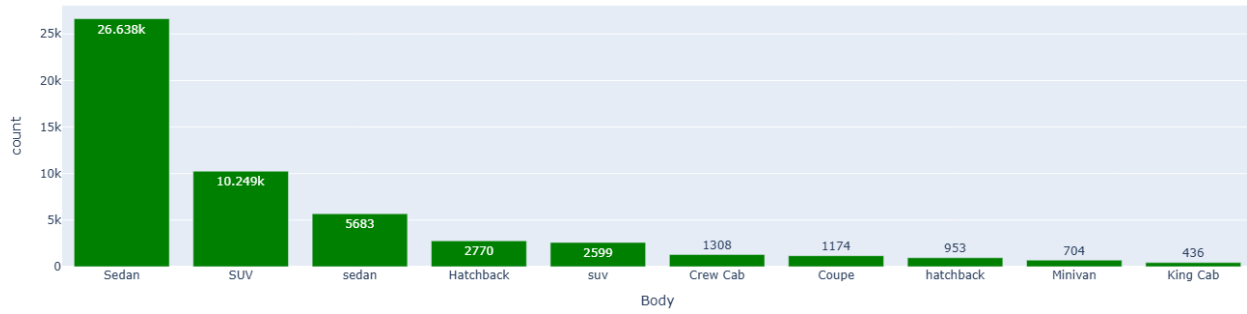


Chevrolet Models

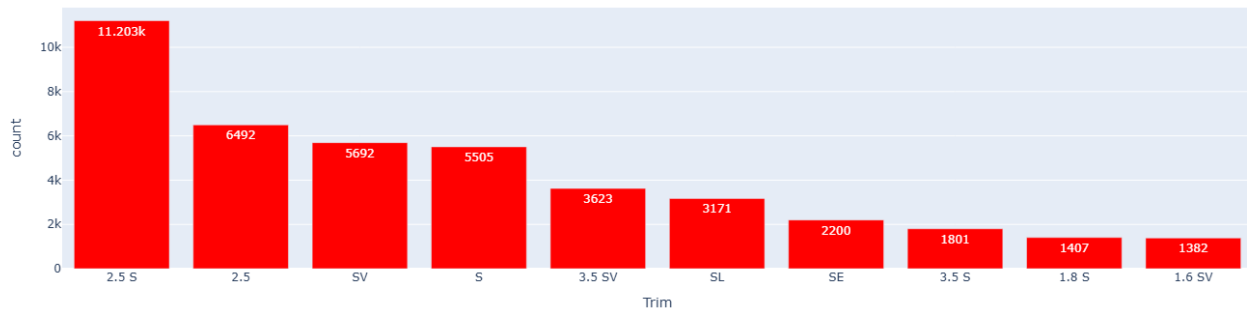


Insight: Silverado and Equinox anchor Chevrolet's performance. The pattern mirrors Ford — Trucks and SUVs dominate, reinforcing that this is a market-wide preference, not brand-specific.

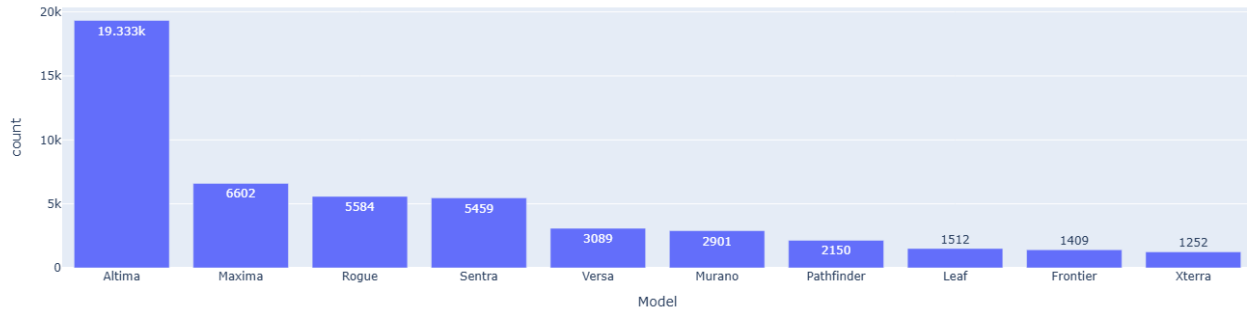
Nissan Body



Nissan Trim

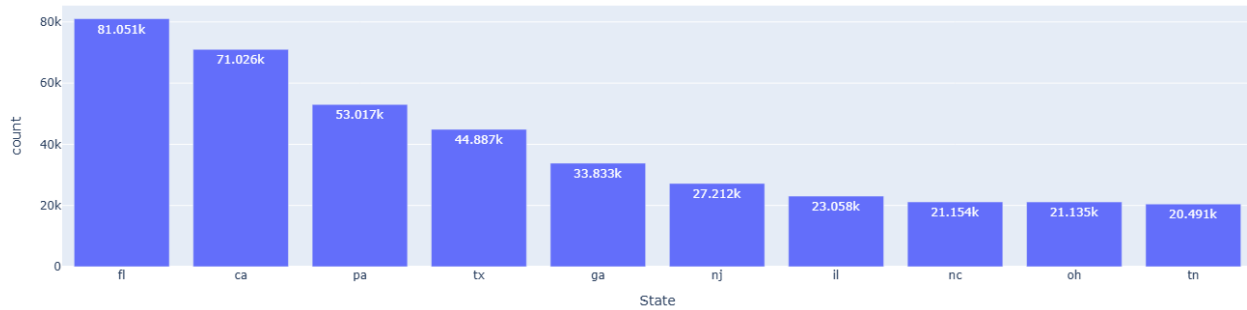


Nissan Models



Insight: Nissan shows a distinctly different buyer profile. The Altima and Sentra — both sedans — top the charts, indicating Nissan attracts a more economy-focused segment.

Top 10 States



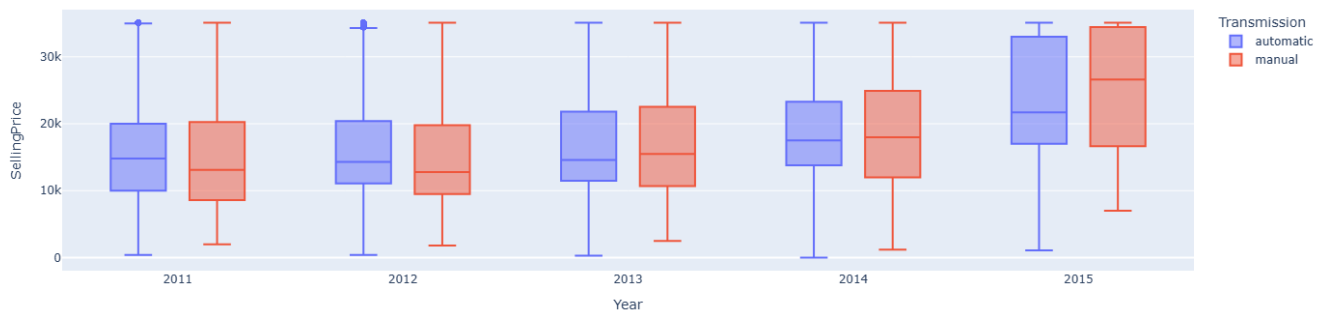
Insight: California, Florida, and Texas account for the largest share of transactions. These three markets offer the highest sales velocity and should be prioritized in any regional strategy.

Top 8 Colores



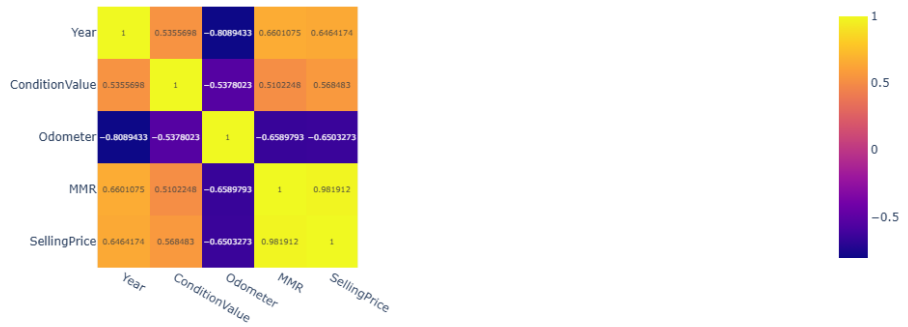
Insight: White, black, and silver account for the overwhelming majority of sales. Unusual or bold colors create friction in the resale process and should be avoided when acquiring inventory.

Automatic vs Manual

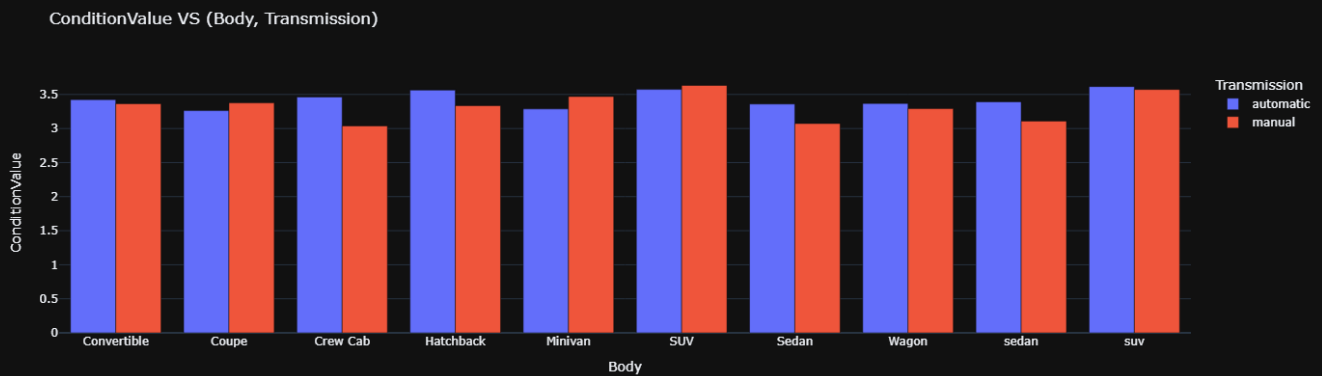


Insight: Automatic vehicles command higher prices and sell in far greater numbers. Manual transmission is a niche preference with measurably lower demand and lower average selling prices.

Correlation Heatmap

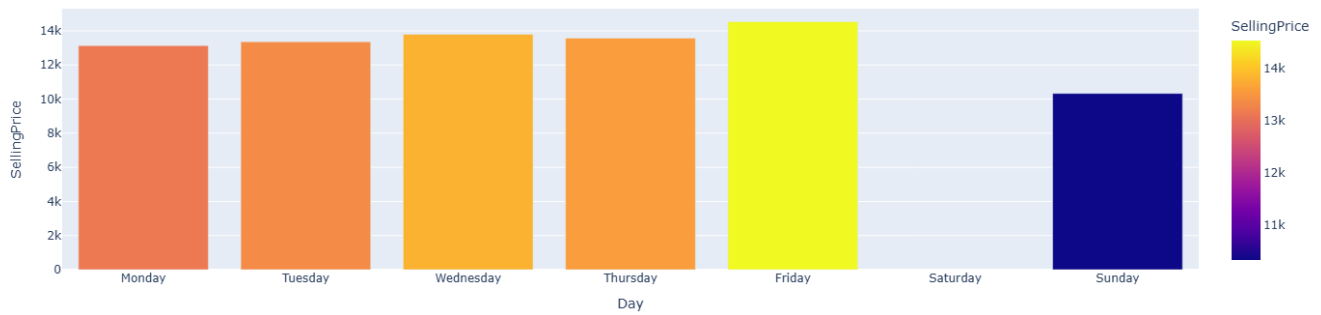


Insight: MMR (Manheim Market Report) is the single strongest predictor of final selling price — more reliable than condition, odometer, or year alone. It should anchor every pricing decision.



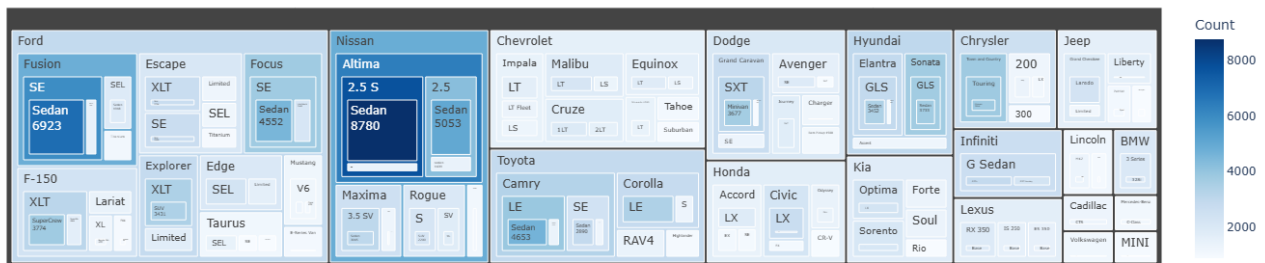
Insight: Condition scores vary significantly across body types and transmission types. Knowing the expected condition range for a given vehicle type allows more accurate pre-purchase quality estimates.

Avg (SellingPrice VS SaleDate)



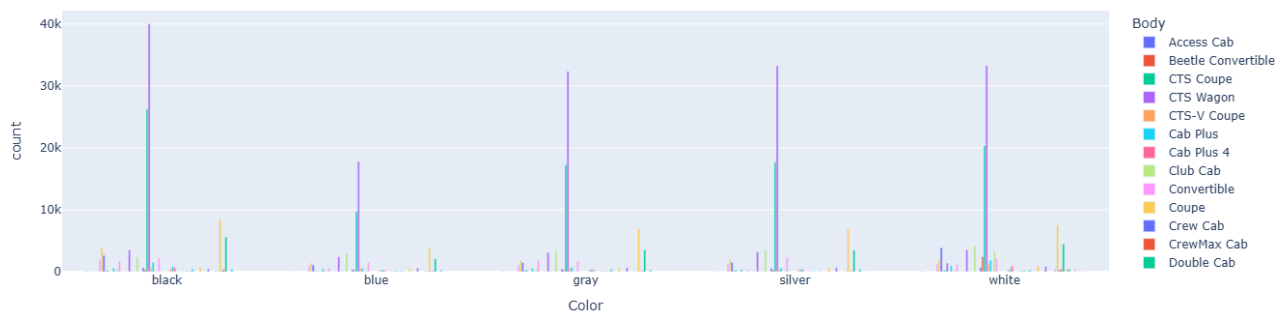
Insight: Selling price averages differ across the days of the week. Identifying peak-price days creates a simple, actionable level to improve per-unit revenue with no additional cost.

Most requested specifications

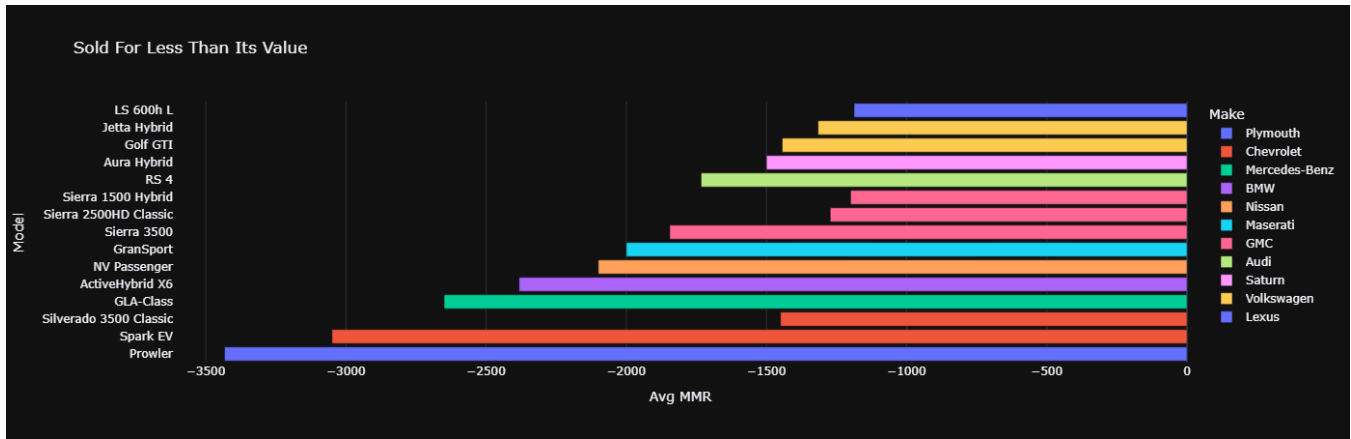


Insight: Specific Make + Model + Trim + Body combinations dominate demand. Acquiring inventory that matches these high-frequency combinations dramatically reduces days-to-sale.

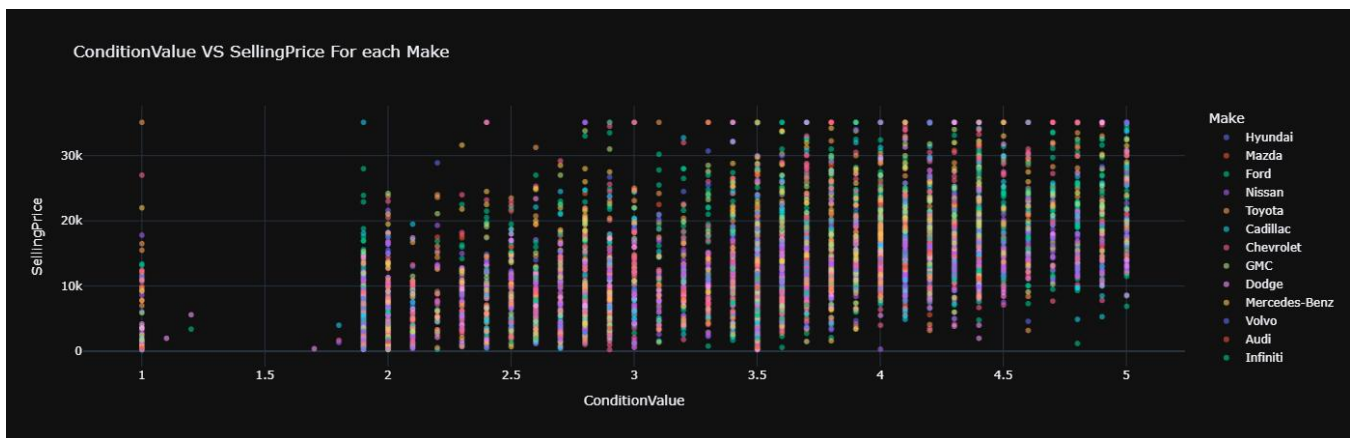
Color VS Body



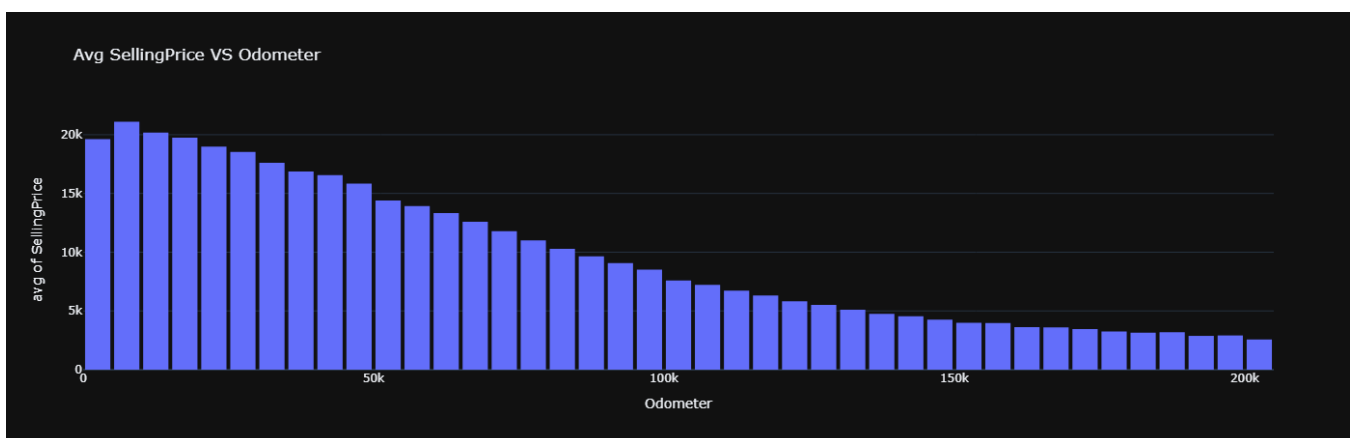
Insight: Neutral-colored SUVs and sedans consistently outperform other combinations. The interaction between color and body type is a stronger demand signal than either factor alone.



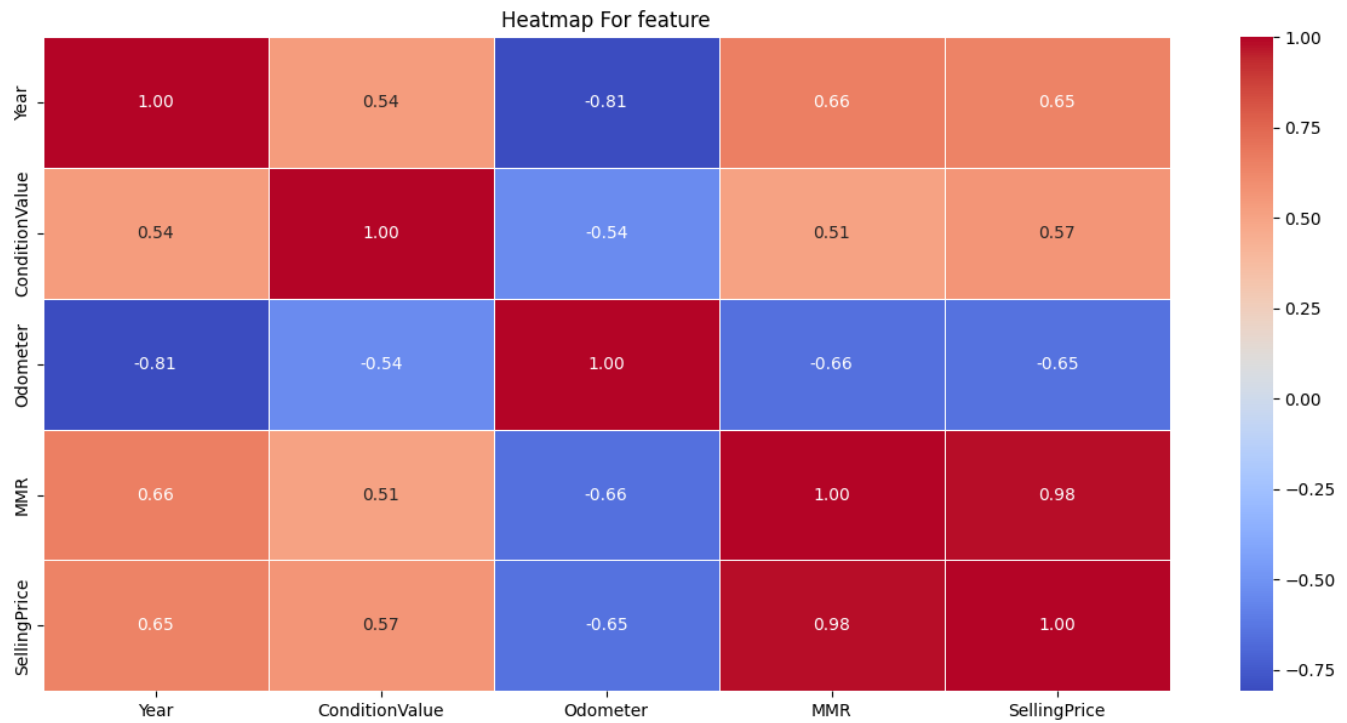
Insight: A meaningful subset of transactions closed below the MMR benchmark. These represent clear buying opportunities — and a warning for sellers who underprice without realizing it.



Insight: Certain premium brands sustain elevated prices even when condition is below average. Brand equity is a real pricing buffer that partially decouples price from physical state.

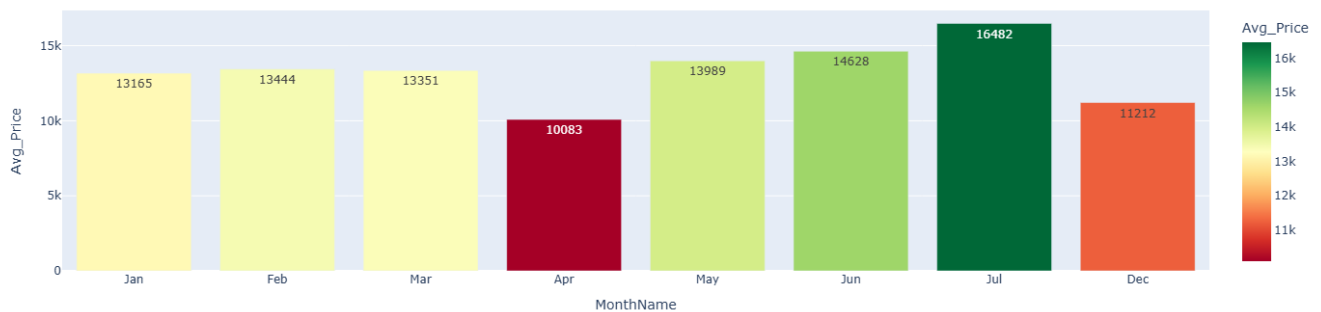


Insight: Price declines predictably with mileage up to a threshold — after which it flattens. Even high-mileage vehicles retain a minimum floor value, which limits downside risk.



Insight: Model year and MMR together explain the largest share of price variation. Other variables contribute, but these two dominate. Any pricing model should treat them as primary inputs.

Avg SellingPrice For each Month



Insight: The market follows a seasonal rhythm. Prices soften in early months like April — the optimal time to buy. They peak mid-year around July — the optimal time to sell.

Chart 21 — Price Outliers

[Insert Chart Here]

Insight: Near-new vehicles are increasingly attracting prices well above their MMR. This signals strong buyer willingness to pay a premium for vehicles that feel brand-new.

Chart 22 — Best Profit Opportunities

[Insert Chart Here]

Insight: The Hummer H2 SUT shows the largest gap between market value and actual selling price — making it an outstanding candidate for targeted profit maximization.

Chart 23 — Condition vs Selling Price

[Insert Chart Here]

Insight: Vehicles in near-perfect condition consistently sell at a meaningful premium. Investing in vehicle presentation and maintenance directly translates into higher realized prices.

Final Conclusion

The U.S. vehicle auction market rewards specific, predictable choices. Newer automatic vehicles in neutral colors from Ford, Chevrolet, or Nissan — purchased at or below MMR in high-volume states — consistently outperform. MMR remains the single most reliable pricing anchor. Condition, seasonality, and body type are the key secondary levers. Applying these insights systematically turns data into a durable competitive advantage.

03 Recommendations for Further Analysis

Monthly & Quarterly Trend Analysis

Expand the time dimension beyond day-of-week. A full seasonality model across months and quarters will reveal when to buy, when to hold, and when to sell — turning timing into a systematic advantage rather than guesswork.

Seller Performance Analysis

The Seller column is currently underutilized. Segmenting by seller type — fleet, lease return, dealer, rental — and comparing their average selling prices relative to MMR could expose which channels consistently over- or under-price.

Price Prediction Model

Build a regression or gradient boosting model using MMR, ConditionValue, Odometer, Year, Make, Body, and State as features. Even a basic model trained on this dataset should achieve strong predictive accuracy for SellingPrice.

Color Premium Analysis

Go beyond color popularity counts — calculate average SellingPrice and average days-to-sale per color. The hypothesis is that white and black don't just sell faster — they may command a measurable price premium worth quantifying.

Geographic Price Arbitrage

The same make, model, trim, and year can sell for thousands more in certain states. A systematic cross-state price comparison matrix would identify specific arbitrage routes worth acting on.

Condition Depreciation by Brand

Model how ConditionValue decays as Odometer increases — and whether this decay rate differs across brands. Brands with more durable condition curves represent better long-term inventory bets.